

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Architected and operated **large-scale data ingestion pipelines** on **GCP** using **Terraform** Infrastructure-as-Code to manage cloud infrastructure for enterprise applications, reducing deployment time by **40%** and enabling cost-efficient data collection at petabyte scale.
- Developed **Python scripts and bash automation** in **Linux** environments to discover, ingest, and process multiple audio data sources into unified pipelines, increasing data acquisition throughput by **3.5x** while maintaining data quality standards.
- Built and scaled **web crawlers** and **large-scale data processing workflows** to extract structured datasets from heterogeneous sources, integrating **Docker** containerization for reproducible pipeline execution across **GCP** infrastructure.
- Collaborated with ML research scientists and data engineers to optimize the **cost/throughput/quality frontier** of data collection operations, delivering **15% cost reduction** and **25% quality improvement** in training datasets through infrastructure optimization and workflow refinement.
- Managed complex data pipeline orchestration and ETL workflows using **Python** and **SQL**, implementing automated data validation and transformation logic to ensure dataset integrity for model training at scale.
- Led cross-functional coordination with AI Team leadership to define and execute the dataset roadmap, translating research requirements into infrastructure improvements and managing priorities across multiple competing data-collection initiatives.

SKILLS

Cloud & Infrastructure – GCP (Compute Engine, Cloud Storage, BigQuery), Terraform, Infrastructure-as-Code, Docker containerization.

Data Engineering & Pipelines – Python scripting, bash, web crawlers, large-scale data processing workflows, ETL, data ingestion pipelines, data validation.

Programming & Scripting – Python, bash, shell scripting, Linux environments, SQL (PostgreSQL, MySQL, Oracle).

DevOps & Release Management – CI/CD pipelines, Git, change-control management, infrastructure automation, deployment orchestration.

Data Management – Data pipelines, data warehousing, schema design, dataset curation, petabyte-scale data operations.

Collaboration & Communication – Cross-functional teamwork, research scientist partnerships, requirements gathering, technical documentation, asynchronous communication.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Petabyte-Scale Audio Data Ingestion & Infrastructure Platform (HCL Technologies)

- Designed and deployed a **Terraform-managed GCP infrastructure** supporting multi-source audio data ingestion, web crawling, and processing workflows; reduced infrastructure provisioning time from **3 weeks to 2 days** and cut operational costs by **32%** through automated resource optimization.
- Engineered **Python and bash automation scripts** to discover, validate, and ingest audio datasets from **15+ heterogeneous data sources** (APIs, web archives, streaming platforms, public datasets) into unified cloud pipelines, achieving **4.2 petabytes ingested monthly** with **99.2% data integrity**.
- Implemented **Docker containerization** for all pipeline components, enabling reproducible builds and **5x faster scaling** during peak ingestion periods; managed container orchestration across **GCP Compute Engine** with automated health checks and recovery.
- Built **data quality frameworks** using **Python and SQL** to validate schema conformance, detect anomalies, and flag corrupt records before model training; reduced data-quality incidents by **68%** and eliminated downstream ML training failures.
- Collaborated with ML scientists to identify cost/quality trade-offs in data sourcing; reduced per-gigabyte collection cost by **28%** while improving training dataset diversity, directly enabling **2.3% improvement** in model accuracy metrics across audio synthesis tasks.

Multi-Source Data Acquisition & Pipeline Orchestration System (HCL Technologies)

- Led requirements gathering and roadmap design with AI Team leadership and data scientists to prioritize **15 competing data sources** and align infrastructure investment with model training priorities; established data SLAs and delivery cadences resulting in **100% on-time dataset delivery**.
- Implemented **large-scale data processing workflows** using **Python pandas, NumPy, and custom ETL logic** to normalize, deduplicate, and enrich audio datasets from **web crawlers, public archives, and proprietary APIs**; processed **2.8 petabytes of raw data monthly** with **99.8% correctness**.
- Automated infrastructure monitoring and cost tracking using **GCP Cloud Logging and custom Python dashboards**; identified inefficiencies and reallocated compute resources, reducing monthly cloud spend by **\$18K** without sacrificing throughput or quality.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)